Taller: Integrales Triples

Integral De Una Constante:

$f(x,y,z) = 1$ Limites: $x \in [0,1]$ $y \in [0,2]$
$z \in [0,3]$

$\int_0^3 1\, \partial z = (z)_0^3 = 3$ $\int_0^2 3\, \partial y = 3y\big|_0^2 = 6$

$\int_0^1 6\, \partial x = 6x\big|_0^1 = 6$ //

Polinomica:

$f(x,y,z) = x+y+z$ Limites: $x \in [0,1]$, $y \in [0,1]$,
$z \in [0,1]$

$\int_0^1 (x+y+z)\, \partial z = (x+y)z + \frac{z^2}{2} = x+y+\frac{1}{2}$

$\int_0^1 \left(x+y+\frac{1}{2}\right) \partial y = xy + \frac{y^2}{2} + \frac{y}{2}$

$= x + \frac{1}{2} + \frac{1}{2} = x+1$

$\int_0^1 (x+1)\, \partial x = \frac{x^2}{2} + x = \frac{1}{2} + 1 = \frac{3}{2}$ //

Separada:

$f(x,y,z) = xy^2z^3$ Limites: $x \in [0,1]$, $y \in [0,2]$
$z \in [0,3]$

$\int_0^3 xy^2z^3\, \partial z = xy^2 \frac{z^4}{4} = xy^2 \cdot \frac{81}{4}$

$\int_0^2 \frac{81}{4} xy^2\, \partial y = \frac{81}{4} x \cdot \frac{y^3}{3} = \frac{81}{4} x \cdot \frac{8}{3} = 54x$

$\int_0^1 54x\,dx = 27$

Exponencial Sencilla

$f(x,y,z) = e^x + y + z$; límites: $x \in [0,1]$, $y \in [0,1]$, $z \in [0,1]$

$\int_0^1 (e^x + y + z)\,dz = (e^x + y)z + \frac{z^2}{2}$

$= e^x + y + \frac{1}{2}$

$\int_0^1 (e^x + y + \frac{1}{2})\,dy = e^x + 1$

$\int_0^1 (e^x + 1)\,dx = (e - 1) + 1 = e$

Multiplicación Basica:

$f(x,y,z) = 2xyz$ Limites: $x \in [0,2]$, $y \in [0,1]$, $z \in [0,2]$

$\int_0^2 2xyz\,dz = 2xy\frac{z^2}{2} = 2xy$

$\int_0^1 2xy\,dy = x$ $\quad \int_0^2 x\,dx = 2$

• Jesús David Méndez